Hilbert's Nullstellensatz

(Proof of existence of transcendental basis is taken from Field and Galois theory Patrick Morandi)

Definition (transcendence basis)

If E/F is a field extension, than a transcendence basis for the extension is a set $S \subseteq E$ such that

- all the elements of S are algebraically independent over F;
- $E/F(S)$ is an algebraic extension.

Lemma.

Suppose that E/F is a field extension such that $S \subseteq E$ is set of algebraically independent elements over F.
If $t \in E$ is transcendental over $F(S)$, then $S \cup \{t\}$ is also algebraically independent over F.

Proof.
Suppose lemma is false.
Then there exists a polynomial p with coefficients in F such that $p(S, t) = 0$. t must appear to a nonzero power as S is algebraically independent, so write $p(S, t) = \sum_{i=0}^{n} p_i(S)t^i = 0$. But then t is not transcendental over $F(S)$. ■

Theorem. (Existence of transcendence basis)

Every field extension E/F admits a transcendence basis.

Proof.
Let S denote the set of all algebraically independent subsets of E. By Zorn's lemma S has to have a maximal element M. Then by previous lemma E has to be algebraic over $F(M)$, so S is a transcendence basis. ■

Nullstellensatz v1 / Zariski's lemma

Suppose that E/F is a field extension such that E is a finitely generated F-algebra.
Then $[E : F] < \infty$.

Proof.
WLOG let $E = F(x_1, \ldots, x_n)$ such that $x_1, \ldots, x_n$ is a transcendence basis for E/F.
On the other hand, as E is a finitely generated F-algebra we can write $E = F[y_1, \ldots, y_r]$.
Then for each i, $y_i = \frac{f_i}{g_i}$ for some $f_i, g_i \in F[x_1, \ldots, x_n]$. Let $h := g_1 \cdots g_r$, such that $E = F[y_1, \ldots, y_r] \subseteq S_h^{-1} F[x_1, \ldots, x_n]$.
Now as $F[x_1, \ldots, x_n]$ is a UFD, let p be some polynomial in $F[x_1, \ldots, x_n] \setminus F$ that is coprime

to h.
Then $\frac{1}{p} \in E \subseteq S_h^{-1}F[x_1, \ldots, x_n]$, so $\frac{1}{p} = \frac{q}{h^k}$ for some $q \in F[x_1, \ldots, x_n]$ and $k \geq 0$.
But then $h^k = pq$ such that we get a contradiction. Hence the transcendence basis is empty and E/F is algebraic. As it is finitely generated, the degree must then be finite. ■

Nullstellensatz v2

Suppose that k is algebraically closed with $\mathfrak{a} \lhd k[x_1, \ldots, x_n]$ a proper ideal.
Then $V_\mathfrak{a}(k) \to \mathrm{Max}(\frac{k[x_1,\ldots,x_n]}{\mathfrak{a}}) = \mathrm{Max}(k[x_1, \ldots, x_n]) \cap V(\mathfrak{a})$, where $\vec{p} \mapsto \ker(\mathrm{ev}_{\vec{p}})$ defines a bijection.
($V_\mathfrak{a}(k)$ denotes the set of common zeros on $\mathfrak{a}$, $V(\mathfrak{a}) := \{\mathfrak{p} \in \mathrm{Spec}(A) \mid \mathfrak{p} \mid \mathfrak{a}\}$)

Proof.
The well-definedness and injectivity of the map is clear.
Now let $\mathfrak{m} \in \mathrm{Max}(\frac{k[x_1,\ldots,x_n]}{\mathfrak{a}})$.
Then by the correspondence theorem $\mathfrak{m} = \frac{\tilde{\mathfrak{m}}}{\mathfrak{a}}$ where $\mathfrak{a} \subseteq \tilde{\mathfrak{m}} \in \mathrm{Max}(k[x_1, \ldots, x_n])$.
Hence $E := \frac{k[x_1,\ldots,x_n]}{\tilde{\mathfrak{m}}}$ is a field extension of k such that E is a finitely generated k-algebra.
Hence by Theorem 17.1. (Nullstellensatz v1), E/k is a finite extension.
But then k is algebraically closed, such that $E = k$.
Thus for all i there exists $c_i \in k$ such that $x_i + \tilde{\mathfrak{m}} = c_i + \tilde{\mathfrak{m}}$ and $\langle x_i - c_i \rangle \subseteq \tilde{\mathfrak{m}}$, such that $\tilde{\mathfrak{m}} = \ker \mathrm{ev}_{(c_1,\ldots,c_n)}$. ■

Nullstellensatz v3

Suppose that k is algebraically closed, with $\mathfrak{a} \lhd k[x_1, \ldots, x_n]$ a proper ideal.
Then $V_\mathfrak{a}(k) \neq \varnothing$.

Proof.
As $\mathfrak{a}$ is a proper ideal, let $\mathfrak{m}$ be a maximal ideal that contains $\mathfrak{a}$.
Then by Theorem 17.2. (Nullstellensatz v2), $\mathfrak{m} = \ker \mathrm{ev}_{\vec{p}}$, such that $\vec{p} \in V_\mathfrak{a}(k)$. ■

Nullstellensatz v4

Suppose that k is algebraically closed.
Then $I(V_\mathfrak{a}(k)) = \sqrt{\mathfrak{a}}$.

Proof.
$(\supseteq)$: Clear.
$(\subseteq)$: Suppose for a contradiction that there exists an $f \in I(V_\mathfrak{a}(k)) \setminus \sqrt{\mathfrak{a}}$.
Then $\{1, f, f^2, \ldots\} \cap \mathfrak{a} = \varnothing$, such that $S_f^{-1}\mathfrak{a} \lhd S_f^{-1}k[x_1, \ldots, x_n] \simeq \frac{k[x_1,\ldots,x_n,y]}{\langle yf-1 \rangle}$ is a proper ideal.
So $\tilde{\mathfrak{a}} := \frac{\mathfrak{a}[y]+\langle yf-1 \rangle}{\langle yf-1 \rangle} \lhd \frac{k[x_1,\ldots,x_n,y]}{\langle yf-1 \rangle}$ is a proper ideal as well.
Hence by Theorem 17.3. (Nullstellensatz v3), $V_{\tilde{\mathfrak{a}}}(k)$ has a point $(\vec{p}_0, y_0)$.
But then $\vec{p}_0 \in V_\mathfrak{a}(k)$, such that $f(\vec{p}_0) = 0$ and $f(\vec{p}_0)y_0 = 1$ at the same time; which is a contradiction. ■

Remark.

All versions of the Nullstellensatz as given above are equivalent.

Proof.

- $(4) \to (2)$: Suppose $\mathfrak{m} \in \operatorname{Max}(k[x_1, \ldots, x_n])$. Then $I(V_{\mathfrak{m}}(k)) = \sqrt{\mathfrak{m}} = \mathfrak{m}$. Then $V_{\mathfrak{m}}(k) \neq \varnothing$ (because inclusion reversing means $I(\varnothing)$ has to be contained in every ideal) so it has to have a point.
- $(2) \to (1)$: Suppose that E/F is a field extension such that E is a finitely generated F-algebra.
  Write $E = \frac{F[\ldots]}{\mathfrak{m}}$ with $\mathfrak{m}$ maximal, so E embeds into $\frac{\overline{F}[\ldots]}{\mathfrak{m}^e} = \overline{F}$ (why is $\mathfrak{m}^e$ a maximal ideal?). So E/F is algebraic. ■